

Polarization dynamics of a two-photon laser

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The high degree of temporal coherence of laser light arises from a complex interplay between the fundamental light-matter interactions of absorption, spontaneous emission, and stimulated emission. The coherence properties of the generated light can be altered significantly, and often in a surprising manner, by modifying the type of light-matter interaction on which the laser is based. As an example, the two-photon laser [1, 2] is based on the higher-order two-photon stimulated emission process, whereby two incident photons stimulate an excited atom to a lower energy state and four photons are scattered, as shown schematically in Fig. 1. While replacing the standard one-photon stimulated emission process by a high-order one might be expected to give rise to subtle differences observable only at the quantum level, it has been predicted that there will be dramatic changes in both the microscopic and macroscopic laser behavior even when many atoms participate in the lasing process. The reason for these differences is that the two-photon stimulated emission rate depends quadratically on the incident photon flux, resulting in an inherently nonlinear light-matter interaction.

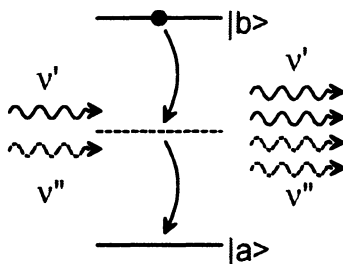


Fig. 1. The idealized two-photon stimulated emission process.

We have recently achieved degenerate two-photon optical lasing [3] using laser-driven potassium atoms contained in a very high finesse optical resonator. This new type of quantum oscillator displays the expected turn-on behavior whereby it can only turn on when an injected coherent field brings the intercavity light above a critical value. Once the laser is above threshold, we observe polarization instabilities and chaos even though the total intensity remains nearly constant. This observation is not encompassed within the framework of current theoretical models of two-photon lasers and results from the atomic Zeeman degeneracy of the states participating in the two-photon stimulated emission process. We suggest that such a two-photon laser could emit polarization-entangled twin beams of light if this degeneracy is lifted, which may have quantum information applications.

To achieve two-photon optical lasing, we use a two-photon gain medium consists of a dense, collimated, effusive beam of laser-driven ^{39}K atoms that passes through the center of the optical resonator. We use the D_1 transition of ^{39}K (6 MHz natural linewidth) whose atomic states are denoted by $|\alpha F_\alpha M_\alpha\rangle$, with $\alpha = g$ for $4^2S_{1/2}$ and $\alpha = e$ for $4^2P_{1/2}$ levels, and F and M are the quantum numbers for the total angular momentum

and its projection on the quantization axis $\hat{z}$. See the energy level diagrams shown in Fig. 2. The ground-state hyperfine splitting is equal to 462 MHz and is denoted by Δ_g . To create a population inversion for multiphoton transitions starting from the state $|g22\rangle$, the atoms are continuously optically pumped into this state by two $\hat{\sigma}_+$ -polarized fields whose frequencies are set close to ω_{21} and ω_{22} (Bohr frequencies are denoted by $\omega_{F_e F_g}$). Note that the atoms are decoupled from the optical pumping fields when they are in this state. The maximum atomic number density in the interaction region is $\sim 2 \times 10^{11}$ atoms/cm³, the atomic beam diameter is 2.5 mm, and the measured residual Doppler absorption width is 30 MHz.

Two-photon stimulated emission into the laser cavity mode (frequency ω_l) occurs when the optically pumped atoms are illuminated by an intense $\hat{\sigma}_-$ -polarized Raman pump beam of frequency ω_d , and ω_l is adjusted to the two-photon transition frequency ($\simeq \omega_d + \Delta_g/2$). Photons are added two at a time to the cavity field via stimulated emission when two photons are annihilated from the $\hat{\sigma}_-$ -polarized Raman pump field and the atom undergoes a multi-photon transition from the state $|g22\rangle$ to any one of the states $|g1M_g\rangle$, as shown schematically in Fig. 2. Note that such a hyper-Raman process can be mapped onto the idealized two-photon stimulated emission process shown in Fig. 1 by working in a dressed-state basis. The atomic, Raman pump beam, and optical cavity axes are mutually orthogonal, thereby suppressing competing phase-matched processes that might give rise to parametric oscillation.

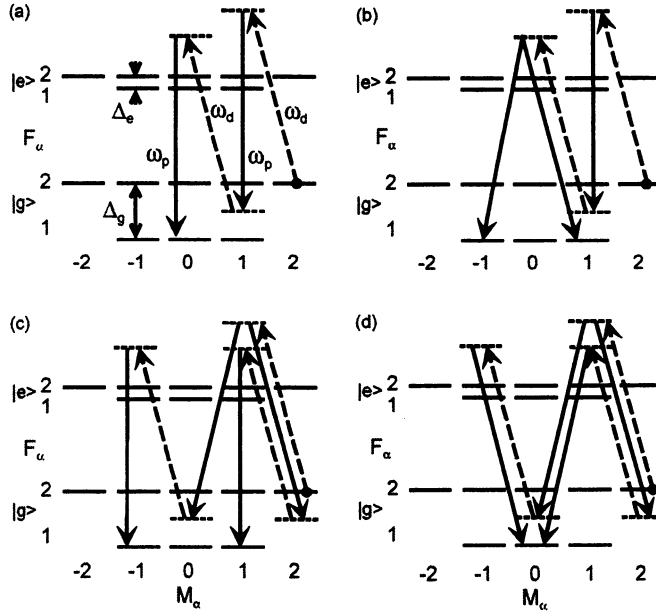


Fig. 2. Two-photon Raman scattering diagrams showing the quantum pathways for the possible states of polarization of the two-photon laser field (solid line) and the circularly polarized Raman pump beam (dashed line).

Two-photon amplification

Before discussing the two-photon laser properties, we demonstrate that this system can give rise to two-photon amplification. We characterize the gain medium by measuring the amplification of an intense $\hat{z}$ -polarized probe beam (frequency ω_p) in the absence of the resonator [4]. A typical probe-beam transmission spectrum is displayed in Fig. 3 where the probe- and Raman pump-beam intensities have been made

strong enough to induce the multi-photon process shown in Fig. 2(a). The Raman pump-beam intensity is $\sim 25 \text{ W/cm}^2$, the atomic number density is $\sim 10^{11} \text{ atoms/cm}^3$, and $\omega_d - \omega_{22} = 522 \text{ MHz}$. We observe several stimulated emission features that disappear when the optical pumping beams are blocked. We identify the mechanisms responsible for each gain and absorption feature by their location and probe-power dependence. The lowest-order contributions to these processes are diagrammed as insets to Fig. 3. The two strongest gain peaks arise from one-photon Raman scattering: (α) at $\omega_p = \omega_d$ and (δ) at $\omega_p = \omega_d + \Delta_g$. The two-photon gain feature (γ) is located about halfway between (α) and (δ), ($\omega_p = \omega_d + \Delta_g/2$ in lowest order). The doublet (ϵ) is “pump-dressed” one-photon gain. The weaker peaks are one-photon-emitting transitions originating from ground states much less populated than $|g_{22}\rangle$, as well as gain and absorption lines arising from ^{41}K (6.7% abundance).

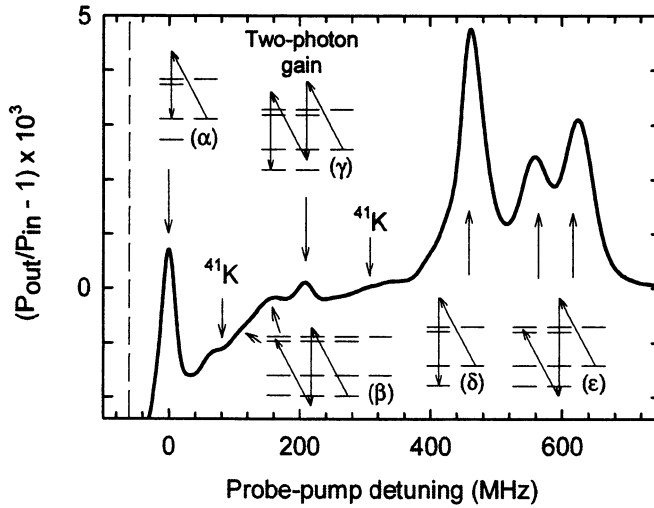


Fig. 3. Intense probe beam transmission spectrum of the Raman-pumped atomic beam.

To demonstrate that the peak denoted by (γ) is a genuine two-photon process, we measure the dependence of its size on the intensity of the probe beam. This intensity dependence is different for one- and two-photon amplification processes. This can be understood simply from an analysis of the amplification process, considering a plane wave propagating through a homogeneously broadened thin medium of length ℓ . The gain experienced by a probe beam and its resonant unsaturated limit are given by

$$G_2 = \frac{I}{I_o} - 1 = \frac{g_2 \ell I_o}{1 + \Delta_2^2 / \Gamma_2^2 + I_o^2 / I_{2s}^2} \rightarrow g_2 \ell I_o, \quad (1)$$

where I (I_o) are the output (input) probe intensities, g_2 is the unsaturated two-photon gain coefficient, I_{2s} is the saturation intensity, Γ_2 is the homogeneous width, and Δ_2 is the detuning from resonance that include the light shifts of the transitions induced by the probe beam. Equation 1 show that the unsaturated two-photon gain is *proportional* to I_o .

Figure 4 displays the height of peak (γ) versus the input probe power. A clear linear increase of the gain is observed for powers up to about 3 mW (24 W/cm² intensity), for which a saturation plateau occurs at a maximum peak height of 4×10^{-4} . This observed linear dependence below saturation is a clear demonstration that photons are

emitted two at a time. The solid line shows the best fit of Eq. 1 to the data with $g_2 = 0.90(6)10^{-4}$ cm/W, and $I_{2s} = 36(3)$ W/cm².

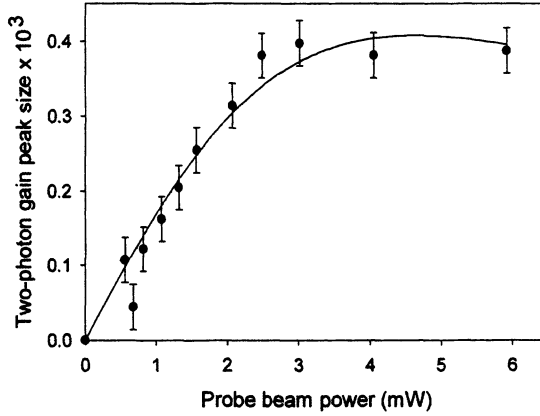


Fig. 4. Size of the two-photon gain peak as a function of the input probe power.

Two-photon lasing

Reaching the threshold for two-photon lasing requires the use of high-quality laser mirrors (developed for cavity quantum electrodynamics experiments) because of the small size of the two-photon gain. The laser resonator is placed in a vacuum chamber (see Fig. 5) and consists of two high-reflectivity ultra-low-loss mirrors of radius of curvature $R=5$ cm set close together at a distance of $L = 1.464$ cm in a linear (standing-wave) transverse-mode-degenerate sub-confocal configuration. This configuration suppresses normal one-photon lasing that can arise from spectrally nearby gain features (one-photon lasing can prevent or obscure two-photon lasing). The laser mirrors are mounted on a super-invar structure to minimize thermal drift of L , with one mirror attached to an electronically controlled mount to permit cavity alignment in vacuum. The other mirror is mounted on a piezoelectric ceramic for fine scanning of the cavity length. The mirrors have a transmissivity of $T = 2 \times 10^{-4}$ as measured by the manufacturer (Research Electro-Optics). We find that the cavity finesse $\mathcal{F} = 15,400 \pm 150 \simeq \pi/(T + A)$ using the ring-down technique, yielding a loss per mirror $A = (4 \pm 2) \times 10^{-6}$.

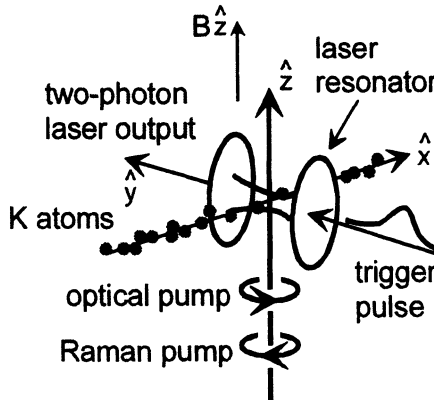


Fig. 5. Two-photon laser experimental setup.

We use a short cavity length so that the free spectral range ($FSR = c/2L = 10.24$

GHz) is larger than the frequency difference between the two-photon gain feature and all other nearby one-photon gain features, which are within 400 MHz of the two-photon frequency. We set the cavity length precisely to a sub-confocal condition $L = R[1 \pm \cos(\pi/p)]$ with $p = 4$ so that the transverse modes group into degenerate clusters, spaced by $c/2pL = 2.56$ GHz and centered on the longitudinal mode frequencies. The spacing between these clusters is also larger than the frequency difference between the two- and one-photon gain features, thus ensuring that one-photon lasing will not occur on a moderate-order transverse mode. Finally, to suppress one-photon lasing on high-order transverse modes (which are no longer frequency degenerate with the lower-order modes due to spherical aberrations), we accurately position under vacuum 400 μm pinholes in front of each cavity mirror and along the cavity axis. With the pinholes in place, $\mathcal{F} = 15,140$, resulting in a longitudinal mode linewidth $FSR/\mathcal{F} = 680$ kHz. Within the cavity mode volume, we estimate that there are a maximum of 7×10^6 atoms. Throughout the experiments, the pump beam frequency is adjusted so that it is tuned to the blue side of the ω_{22} transition by 512 MHz. The mutually orthogonal geometry of our experiment suppresses competing phase-matched processes, ensuring that we are constructing a true two-photon laser and not a parametric oscillator.

Our experimental system is very rich because there are four quantum pathways, depicted in Fig. 2, that connect the initial state $|g_{22}\rangle$ and the final states $|g_1 M_g\rangle$, differ only by the state of polarization of the cavity field, and are frequency degenerate at zero magnetic field. The two-photon resonator can support any state of polarization (it is measured to be highly isotropic) hence all pathways of Fig. 2 must be considered.

Conclusive evidence that we have achieved two-photon lasing is indicated by the threshold behavior of the laser. Briefly, the threshold condition for all lasers is that the round-trip gain must equal the round-trip loss. For one-photon lasers this yields the well known result that lasing will commence when a uniquely defined minimum inversion density (proportional to the gain) is attained via sufficient pumping. The situation is more complicated for the two-photon laser because the unsaturated gain increases with increasing inversion density *and* with increasing cavity photon number. This results in a threshold condition specified by a uniquely defined minimum inversion density and cavity photon number [5]. If the laser is pumped strongly but with no photons in the cavity initially, it cannot turn on unless quantum fluctuations [6] or an injected field brings the intracavity light above the critical value. In addition, once the minimum photon number exists in the cavity, the photon number undergoes a run-away process, growing rapidly until the two-photon transition is saturated [5]. Therefore, the two-photon laser operates in the saturated regime (a source of optical nonlinearity) even at the laser threshold, giving rise to the possibility that the laser will display dynamical instabilities.

To observe the threshold behavior, we measure the power of the beam emitted from the two-photon laser resonator and inject pulses of light into the cavity to trigger the laser to the 'on' state and shown in Fig. 6(a) for $\sim 3.3 \times 10^5$ injected photons. The duration of the injected pulse is indicated by the dashed vertical lines. It is seen that the power emitted by the cavity rises to approximately 0.2 μW and remains at this value, corresponding to an average cavity photon number of 2.2×10^6 and an intracavity intensity of 7.3 W/cm² at the cavity waist.

While the total emitted power is seen to remain nearly constant, we find that the two-photon laser displays polarization instabilities. By placing a linear polarization before the detector, it is seen in Fig. 6(b) that the state of polarization undergoes very regular oscillations of period 0.11 μs , with a 50% depth of modulation. Similar dynamical behavior is observed for all polarizer orientations, suggesting that the state of polarization is elliptical with an ellipticity of 0.5 and a rotating major axis. We find

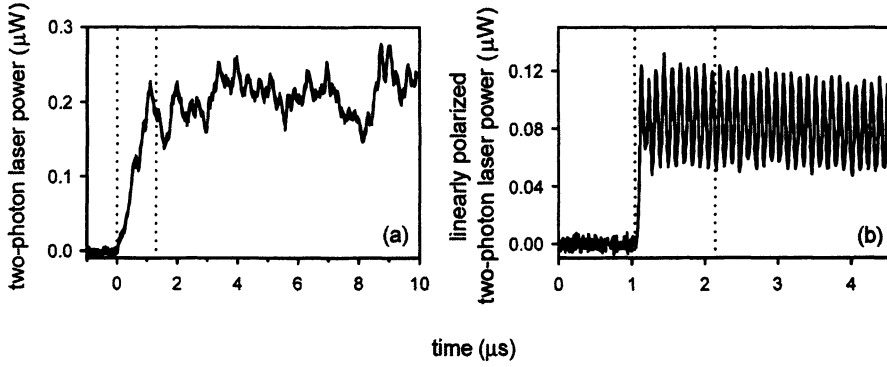


Fig. 6. (a) Turn on behavior of the two-photon laser. (b) Polarization dynamics of the two-photon laser.

that the laser dynamics is quite sensitive to the applied magnetic field: an increase by as little as 0.5 Gauss is sufficient to generate a much more complicated pattern, turning into a possibly chaotic pattern for a field strength of 2 Gauss. We believe that it may be possible to suppress the instabilities using a stronger magnetic field to lift the degeneracy of the different quantum pathways.

Lifting the degeneracy between the pathways also has the promise of yielding polarization entanglement. Consider, for simplicity, a single emission event. When the Zeeman levels are degenerate, the superposition of all possible amplitudes gives a photon pair (1,2) in the state $[\alpha_1|\hat{z}\rangle_1 + \beta_1|\hat{x}\rangle_1][\alpha_2|\hat{z}\rangle_2 + \beta_2|\hat{x}\rangle_2]$ with $|\alpha_i|^2 + |\beta_i|^2 = 1$, where $|\hat{\epsilon}\rangle_i$ ($\epsilon = x, y$) is the polarization-labeled Fock state of the i^{th} photon of the emitted pair. In a strong magnetic field, the $\hat{x}\hat{z}$ and $\hat{z}\hat{x}$ pairs become off-resonant and should stop lasing, creating the polarization-entangled state $\alpha_1\alpha_2|\hat{z}\rangle_1|\hat{z}\rangle_2 + \beta_1\beta_2|\hat{x}\rangle_1|\hat{x}\rangle_2$.

In conclusion, we have described the design and characteristics of a Raman two-photon amplifier and laser. The multilevel nature of the two-photon transition allows interesting effects such as polarization instabilities and possible photon entanglement. This opens up exciting lines of study in nonlinear laser dynamics and quantum optics.

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